Statistics Basics

Question 1: What is the difference between descriptive statistics and inferential statistics? Explain with examples.

Ans :-

Basis	**Descriptive Statistics**	**Inferential Statistics**
Purpose	Describe data	Make predictions
Data Used	Entire dataset	Sample data
Outcome	Summary (facts)	Conclusions (estimates)
Example	Average marks of class	Predicting population behavior

Question 2: What is sampling in statistics? Explain the differences between random and stratified sampling.

Ans :- **Sampling** is the process of selecting a small group (sample) from a larger group (population) to study and draw conclusions. Instead of studying the entire population, we study a sample to save time, cost, and effort.

Differences Between Random and Stratified Sampling

Basis	**Random Sampling**	**Stratified Sampling**
Selection	Completely random	From divided groups
Representation	May miss some groups	Covers all groups
Accuracy	Moderate	Higher

Complexity	Simple	Slightly complex

Question 3: Define mean, median, and mode. Explain why these measures of central tendency are important.

Ans :-

A. Mean (Average) :- The mean is the sum of all values divided by the total number of values.

B. Median :- The median is the middle value when data is arranged in ascending or descending order.

C. Mode :-The mode is the value that occurs most frequently in a dataset

Importance :-

1. Simplify Large Data

They reduce a large dataset into a single representative value.

Example: Instead of listing all marks, we say “average marks = 65”.

2. Help in Comparison

Used to compare different datasets easily.

Example: Comparing average salary of two companies.

3. Support Decision-Making

Useful in business, economics, and research for making data-driven decisions.

Example: Deciding pricing based on average customer spending.

4. Identify Data Patterns

- Mean shows overall trend
- Median shows central position (useful with outliers)
- Mode shows most common value

Question 4: Explain skewness and kurtosis. What does a positive skew imply about the data?

Ans :- **Skewness** measures the asymmetry (lack of symmetry) in the distribution of data.

Kurtosis measures the peakedness or flatness of a distribution.

Positive Skew Imply?

A positive skew :-

- Majority of data values are low or moderate
- Few values are extremely high (outliers)
- The tail extends to the right
- Mean is higher than median

Real-life meaning:

In income data, most people earn average income, but a few earn very high salaries, pulling the average upward.

Question 5: Implement a Python program to compute the mean, median, and mode of

a given list of numbers

numbers = [12, 15, 12, 18, 19, 12, 20, 22, 19, 19, 24, 24, 24, 26, 28]

Ans :-

PYTHON CODE

```
import statistics as st

numbers = [12, 15, 12, 18, 19, 12, 20, 22, 19, 19, 24, 24, 24, 26, 28]


# Mean

mean_value = st.mean(numbers)
```

```
# Median
median_value = st.median(numbers)


# Mode
mode_value = st.multimode(numbers)  # handles multiple modes
print("Mean:", mean_value)
print("Median:", median_value)
print("Mode:", mode_value)
```

Output :-

Mean: 19.6

Median: 19

Mode: [12, 19, 24]

Question 6: Compute the covariance and correlation coefficient between the following two datasets provided as lists in Python:

list_x = [10, 20, 30, 40, 50]

list_y = [15, 25, 35, 45, 60]

Ans :-

PYTHON CODE

```
import numpy as np
list_x = [10, 20, 30, 40, 50]
list_y = [15, 25, 35, 45, 60]
x = np.array(list_x)
y = np.array(list_y)
cov_matrix = np.cov(x, y)
```

```
covariance = cov_matrix[0][1]

correlation = np.corrcoef(x, y)[0][1]

print("Covariance:", covariance)

print("Correlation Coefficient:", correlation)
```

Output :-

Covariance: 275.0

Correlation Coefficient: 0.995893206467704

Question 7: Write a Python script to draw a boxplot for the following numeric list and identify its outliers. Explain the result:

data = [12, 14, 14, 15, 18, 19, 19, 21, 22, 22, 23, 23, 24, 26, 29, 35]

Ans :-

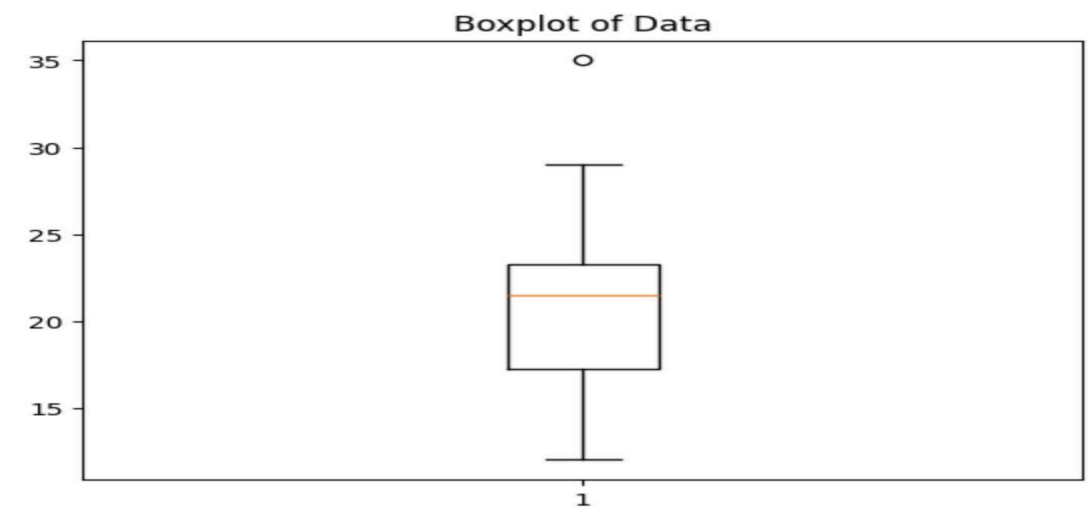
PYTHON CODE

```
import matplotlib.pyplot as plt

import numpy as npm


data = [12, 14, 14, 15, 18, 19, 19, 21, 22, 22, 23, 23, 24, 26, 29, 35]

# Create boxplot

plt.boxplot(data)

plt.title("Boxplot of Data")

plt.show()


# Calculate quartiles

Q1 = np.percentile(data, 25)

Q2 = np.percentile(data, 50)  # Median

Q3 = np.percentile(data, 75)
```

```
# Interquartile Range
IQR = Q3 - Q1

# Outlier boundaries
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

# Identify outliers
outliers = [x for x in data if x < lower_bound or x > upper_bound]

print("Q1:", Q1)
print("Median:", Q2)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower Bound:", lower_bound)
print("Upper Bound:", upper_bound)
print("Outliers:", outliers)
```

Output :-

...




```
Q1: 17.25
Median: 21.5
Q3: 23.25
IQR: 6.0
Lower Bound: 8.25
Upper Bound: 32.25
Outliers: [35]
```

Question 8: You are working as a data analyst in an e-commerce company. The marketing team wants to know if there is a relationship between advertising spend and daily sales.

●Explain how you would use covariance and correlation to explore thisrelationship.

●Write Python code to compute the correlation between the two lists:

advertising_spend = [200, 250, 300, 400, 500]

daily_sales = [2200, 2450, 2750, 3200, 4000]

Ans :- As a data analyst, to examine the relationship between advertising spend and daily sales, I would use both covariance and correlation as follows:

1. Covariance:

Covariance measures the direction of the relationship between two variables.

- If covariance is positive, it indicates that both variables move in the same direction.
- If covariance is negative, they move in opposite directions

In this case, a positive covariance would suggest that an increase in advertising spend leads to an increase in daily sales.

2. Correlation:

Correlation measures both the strength and direction of the relationship. Its value ranges from -1 to +1.

- +1 indicates a perfect positive relationship
- 0 indicates no relationship
- -1 indicates a perfect negative relationship

A correlation close to +1 would imply a strong positive relationship between advertising spend and sales.

Conclusion: By using covariance and correlation together, we can determine whether advertising spend influences sales and how strong that relationship is. This helps the marketing team make better decisions regarding budget allocation.

Python Code

```
import numpy as np

advertising_spend = [200, 250, 300, 400, 500]

daily_sales = [2200, 2450, 2750, 3200, 4000]

# Convert lists to arrays

x = np.array(advertising_spend)

y = np.array(daily_sales)

# Calculate correlation coefficient

correlation = np.corrcoef(x, y)[0][1]

print("Correlation Coefficient:", correlation)
```

Question 9: Your team has collected customer satisfaction survey data on a scale of 1-10 and wants to understand its distribution before launching a new product.

- Explain which summary statistics and visualizations (e.g. mean, standarddeviation, histogram) you'd use.
- Write Python code to create a histogram using Matplotlib for the survey data:

```
survey_scores = [7, 8, 5, 9, 6, 7, 8, 9, 10, 4, 7, 6, 9, 8, 7]
```

Ans :-

To understand the distribution of customer satisfaction scores (1–10 scale), the following methods are useful:

1. Mean (Average)

- Shows the overall satisfaction level of customers
- Helps identify if customers are generally satisfied or not

2. Median

- Represents the middle value
- Useful when data has outliers

3. Standard Deviation

- Measures variability or spread of the data
- Low value → responses are consistent
- High value → responses vary widely

4. Minimum and Maximum

- Show the range of satisfaction scores
- Helps identify extreme values

5. Histogram (Visualization)

- Displays the frequency distribution of scores
- Helps understand:
    - Whether data is skewed or symmetric
    - Where most responses are concentrated

Conclusion

Using these statistics and visualization helps the company understand:

- Overall customer sentiment
- Consistency in responses
- Whether improvements are needed before launching the product

PYTHON CODE

```
import matplotlib.pyplot as plt

survey_scores = [7, 8, 5, 9, 6, 7, 8, 9, 10, 4, 7, 6, 9, 8, 7]

# Create histogram

plt.hist(survey_scores, bins=5)

# Labels and title

plt.xlabel("Survey Scores")

plt.ylabel("Frequency")

plt.title("Customer Satisfaction Distribution")

plt.show()
```